we've seen, found using EPR's hypothesis of a *local* universe). To impressive accuracy, *this is just what Aspect found in his experiments, 50 percent agreement.* Standard quantum mechanics matches the data impressively.

This is a spectacular success. Nevertheless, there is a hitch. *After more than seven decades, no one understands how or even whether the collapse of a probability wave really happens.* Over the years, the *assumption* that probability waves collapse has proven itself a powerful link between the probabilities that quantum theory predicts and the definite outcomes that experiments reveal. But it's an assumption fraught with conundrums. For one thing, the collapse does not emerge from the mathematics of quantum theory; it has to be put in by hand, and there is no agreed-upon or experimentally justified way to do this. For another, how is it possible that by finding an electron in your detector in New York City, you cause the electron's probability wave in the Andromeda galaxy to drop to zero instantaneously? To be sure, once you find the particle in New York City, it definitely won't be found in Andromeda, but what unknown mechanism enforces this with such spectacular efficiency? How, in looser language, does the part of the probability wave in Andromeda, and everywhere else, "know" to drop to zero simultaneously?[19]

We will take up this *quantum mechanical measurement problem* in Chapter 7 (and as we'll see, there are other proposals that avoid the idea of collapsing probability waves entirely), but suffice it here to note that, as we discussed in Chapter 3, something that is simultaneous from one perspective is not simultaneous from another moving perspective. (Remember Itchy and Scratchy setting their clocks on a moving train.) So if a probability wave were to undergo simultaneous collapse across space according to one observer, it will *not* undergo such simultaneous collapse according to another who is in motion. As a matter of fact, depending on their motion, some observers will report that the left photon was measured first, while other observers, equally trustworthy, will report that the right photon was measured first. Hence, even if the idea of collapsing probability waves were correct, there would fail to be an objective truth regarding which measurement—on the left or right photon—affected the other. Thus, the collapse of probability waves would seem to pick out one vantage point as special—the one according